Importing the Dataset into SQL Server

Voltkart · load the data before you start

What you'll do

You'll create a database and load the eight CSV files from the `data/` folder into it. There are two ways: the **Import Flat File wizard** (recommended — it's point-and-click and handles empty values cleanly), or **BULK INSERT** (a scripted alternative). Pick one.

> **Note** This assumes SQL Server and SSMS are already installed and connected. If they aren't,
> follow the "SQL Server & SSMS — Setup Guide" first.

Step 1 — Create the database

1. Open SSMS and connect to your server (for a local install the server name is `localhost`).
2. In a new query window, run:

```
CREATE DATABASE Voltkart;
```

3. In Object Explorer, right-click **Databases** and choose **Refresh**. `Voltkart` now appears in the list.

Step 2, Option A — Import Flat File wizard (recommended)

Do this once per CSV file. The wizard reads the header row, names the table after the file, and infers the column types for you.

1. In Object Explorer, expand **Databases**, right-click **Voltkart**, and choose **Tasks → Import Flat File…**
2. Click **Browse**, pick one CSV from the `data/` folder (for example `fact_orders.csv`), and click **Next**.
3. On the preview screen, confirm the **table name** matches the file name (for example `fact_orders`) and click **Next**.
4. On the **Modify Columns** screen, glance over the detected data types. Leave the defaults unless something is clearly wrong, then click **Next** and **Finish**.
5. Repeat for all eight files: `dim_category`, `dim_employee`, `dim_customer`,

`dim_product`, `fact_orders`, `fact_order_items`, `stg_orders_incr`, `cdc_product_changes`.

> **Tip** The wizard imports the empty values in `parent_category_id` (the category root) and `manager_id` (the CEO) as NULL automatically — exactly what the recursive questions expect. You don't need to do anything special.

Step 2, Option B — BULK INSERT (scripted alternative)

If you prefer to script the load, use the `data/load.sql` helper. It has a `CREATE TABLE` statement and a `BULK INSERT` line for each file.

1. Copy the eight CSV files to a folder the SQL Server service can read, for example `C:\cb_data\`.
2. Open `data/load.sql` in SSMS, make sure it starts with `USE Voltkart;`, and update the file paths in the `BULK INSERT` lines to wherever you put the CSVs.
3. Run the script.

> **Warning** BULK INSERT reads the file from the **server's** file system, not yours, and the SQL Server service account must have permission to that folder. If you hit a permission or "cannot bulk load" error, the Import Flat File wizard in Option A is the simpler route.

Step 3 — Confirm the load

Run this to check the row counts. The numbers on the right are what you should see.

```
USE Voltkart;
SELECT 'dim_category'        AS table_name, COUNT(*) AS rows FROM dim_category
UNION ALL SELECT 'dim_employee',        COUNT(*) FROM dim_employee
UNION ALL SELECT 'dim_customer',        COUNT(*) FROM dim_customer
UNION ALL SELECT 'dim_product',         COUNT(*) FROM dim_product
UNION ALL SELECT 'fact_orders',         COUNT(*) FROM fact_orders
UNION ALL SELECT 'fact_order_items',    COUNT(*) FROM fact_order_items
UNION ALL SELECT 'stg_orders_incr',     COUNT(*) FROM stg_orders_incr
UNION ALL SELECT 'cdc_product_changes', COUNT(*) FROM cdc_product_changes;
```

table_name	expected rows
dim_category	26
dim_employee	58
dim_customer	2,000
dim_product	400
fact_orders	30,000

fact_order_items	56,825
stg_orders_incr	800
cdc_product_changes	35

If every count matches, your environment is ready and you can open the Assignment Questions.

> **Note** Questions 8 and 9 change the data (they run MERGE statements). If you want a clean slate afterwards, just drop the affected tables and re-import them, or re-run the data generator if you have it.